

MetaFrontierLab Benchmark Report

Generated from: results/benchmarks_scaled_full_corrected

Replications per scenario: 5

Methods benchmarked: tbema, exact_baseline, dersimonian_laird, henmi_copas, metafor_trimfill_external, metafor_selmodel_external, copas_selection_external, robma_bibma_external

Executive Summary

- Best overall RMSE in this run: **robma_bibma_external** at 0.463.
- Highest observed coverage: **tbema** at 0.850.
- Fastest method: **dersimonian_laird** at 0.001 seconds per fit.
- Methods with incomplete runs: copas_selection_external, exact_baseline, metafor_selmodel_external.
- Runtime values for external R methods include adapter startup and package-load overhead, so they reflect end-to-end benchmark cost rather than pure algorithm time.

Overall Method Ranking

attempted_runs	successful_runs	invalid_ok_runs	skipped_runs	error_runs	success_rate	bias	mean_absolute_error	rmse	coverage_95	mean_runtime
20	20	0	0	0	1.000	0.306	0.388	0.463	0.650	0.001
20	20	0	0	0	1.000	0.324	0.402	0.478	0.700	0.001
20	20	0	0	0	1.000	0.407	0.485	0.541	0.600	0.001
20	20	0	0	0	1.000	0.044	0.421	0.558	0.850	0.001
20	20	0	0	0	1.000	0.458	0.513	0.566	0.550	0.001
20	19	0	0	1	0.950	0.460	0.536	0.615	0.526	0.001
20	18	0	0	2	0.900	0.182	0.417	0.509	0.611	0.001
20	17	0	0	3	0.850	0.429	0.475	0.580	0.706	0.001

Scenario Highlights

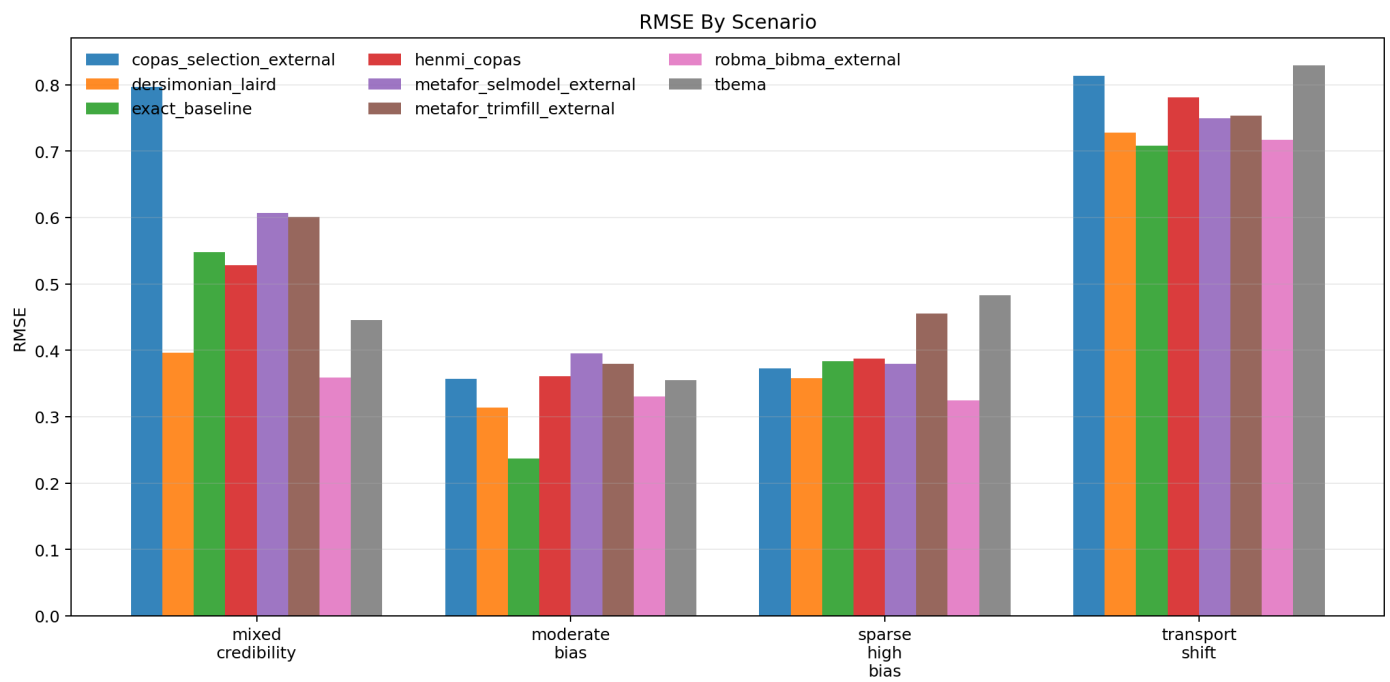
- **mixed_credibility**: best RMSE was robma_bibma_external (0.358); highest coverage was tbema (1.000). Incomplete runs: copas_selection_external, exact_baseline.
- **moderate_bias**: best RMSE was dersimonian_laird (0.314); highest coverage was dersimonian_laird (0.800). Incomplete runs: exact_baseline.
- **sparse_high_bias**: best RMSE was robma_bibma_external (0.324); highest coverage was robma_bibma_external (1.000). Incomplete runs: metafor_selmodel_external.
- **transport_shift**: best RMSE was exact_baseline (0.708); highest coverage was tbema (0.600).

Scenario Table

	replications_total	successful_runs	invalid_ok_runs	skipped_runs	error_runs	success_rate	bias	rmse	coverage_95	mean
ection_external	5	4	0	0	1	0.800	0.678	0.797	0.250	
simonian_laird	5	5	0	0	0	1.000	0.312	0.396	0.800	
exact_baseline	5	4	0	0	1	0.800	-0.194	0.548	0.750	
henmi_copas	5	5	0	0	0	1.000	0.469	0.528	0.600	
odel_external	5	5	0	0	0	1.000	0.482	0.607	0.800	
rimfill_external	5	5	0	0	0	1.000	0.557	0.601	0.400	
tbma_external	5	5	0	0	0	1.000	0.289	0.358	0.800	
tbema	5	5	0	0	0	1.000	-0.095	0.446	1.000	
ection_external	5	5	0	0	0	1.000	0.208	0.357	0.800	
simonian_laird	5	5	0	0	0	1.000	0.141	0.314	0.800	
exact_baseline	5	4	0	0	1	0.800	0.191	0.237	0.750	
henmi_copas	5	5	0	0	0	1.000	0.189	0.361	0.600	
odel_external	5	5	0	0	0	1.000	0.224	0.395	0.800	
rimfill_external	5	5	0	0	0	1.000	0.248	0.380	0.800	
tbma_external	5	5	0	0	0	1.000	0.119	0.330	0.600	
tbema	5	5	0	0	0	1.000	-0.004	0.355	0.800	
ection_external	5	5	0	0	0	1.000	0.206	0.373	1.000	
simonian_laird	5	5	0	0	0	1.000	0.192	0.358	1.000	
exact_baseline	5	5	0	0	0	1.000	0.043	0.383	0.800	
henmi_copas	5	5	0	0	0	1.000	0.231	0.387	1.000	
odel_external	5	2	0	0	3	0.400	0.303	0.380	1.000	
rimfill_external	5	5	0	0	0	1.000	0.324	0.455	0.800	
tbma_external	5	5	0	0	0	1.000	0.180	0.324	1.000	
tbema	5	5	0	0	0	1.000	-0.142	0.483	1.000	
ection_external	5	5	0	0	0	1.000	0.791	0.813	0.000	
simonian_laird	5	5	0	0	0	1.000	0.649	0.728	0.200	
exact_baseline	5	5	0	0	0	1.000	0.615	0.708	0.200	
henmi_copas	5	5	0	0	0	1.000	0.740	0.781	0.200	
odel_external	5	5	0	0	0	1.000	0.630	0.749	0.400	
rimfill_external	5	5	0	0	0	1.000	0.702	0.754	0.200	
tbma_external	5	5	0	0	0	1.000	0.638	0.717	0.200	
tbema	5	5	0	0	0	1.000	0.417	0.829	0.600	

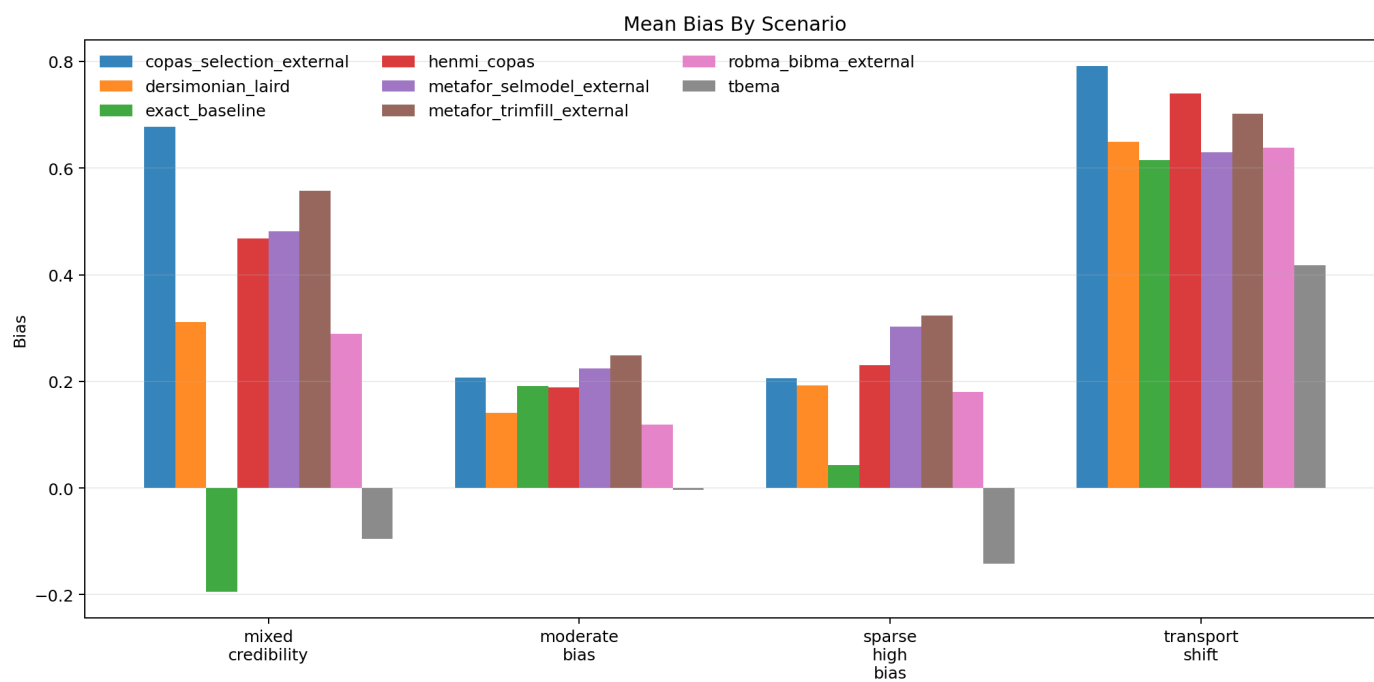
RMSE by Scenario

Figure generated from the benchmark summary.



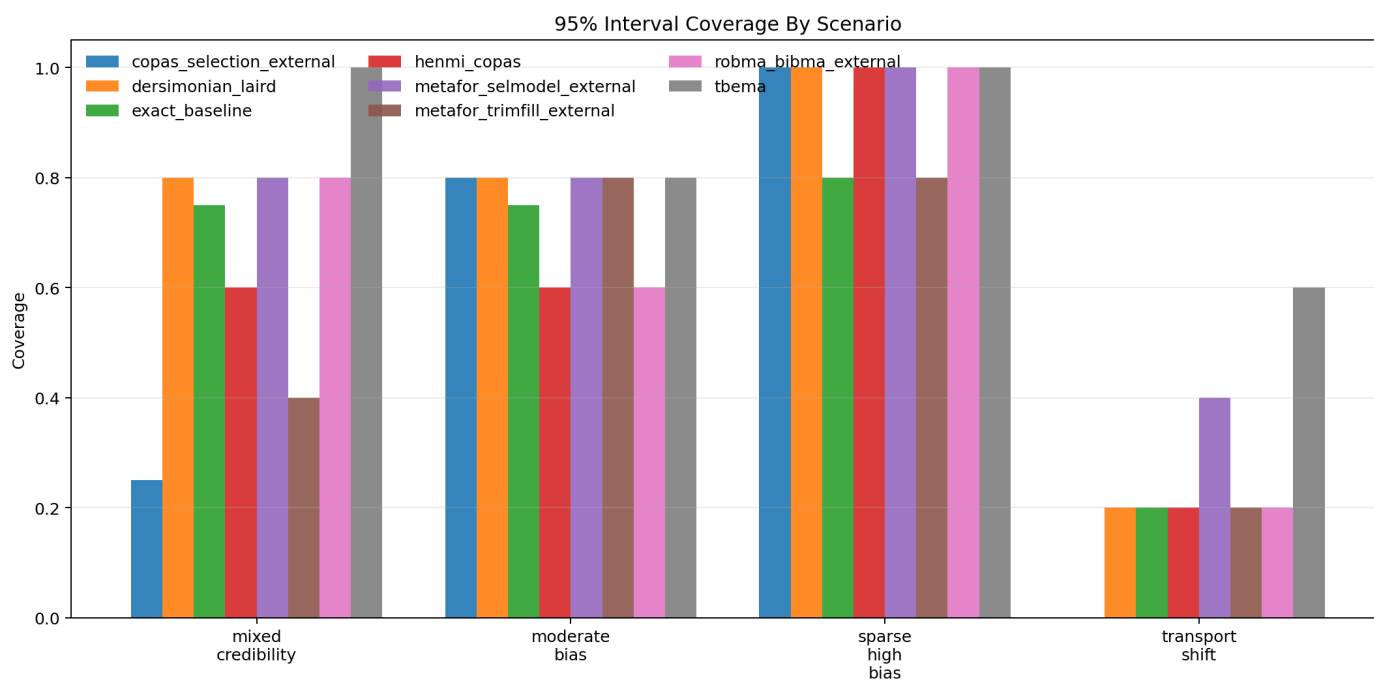
Bias by Scenario

Figure generated from the benchmark summary.



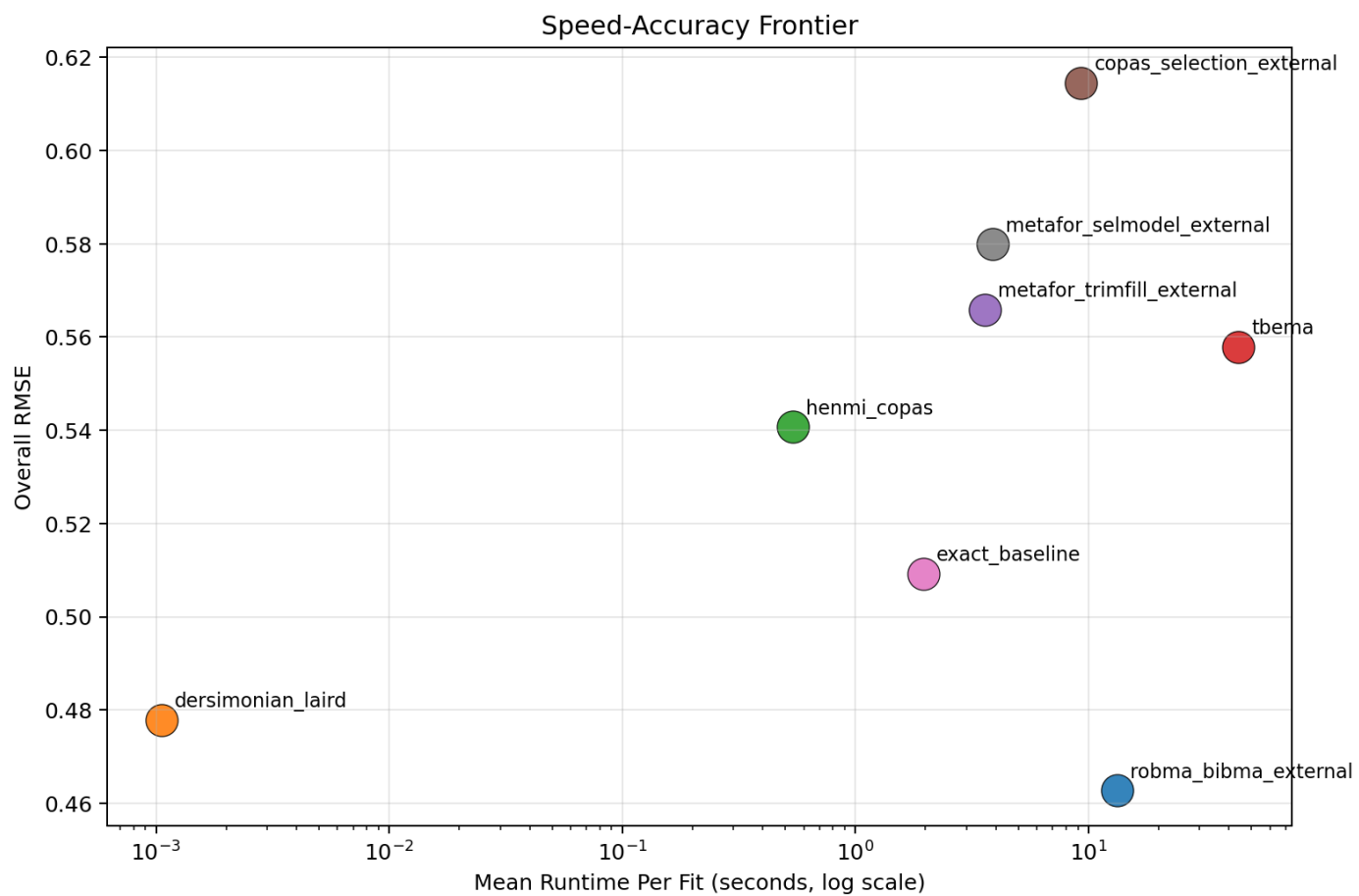
Coverage by Scenario

Figure generated from the benchmark summary.



Speed-Accuracy Frontier

Figure generated from the benchmark summary.



Methods Appendix

- TBEMA: exact sparse-data meta-analysis with stacked bias tempering, transport weighting, and ridge-regularized target-aware meta-regression.
- DerSimonian-Laird: standard normal-normal random-effects benchmark.
- Henmi-Copas: publication-bias-robust interval method.
- metafor trimfill: trim-and-fill adjustment from metafor.
- metafor selmodel: step-function selection model from metafor.
- Copas selection: Copas model from metasens.
- RoBMA BiBMA: Bayesian model averaging for binary outcomes, using the fast spike-and-slab algorithm.